Practice Problems

ORIGINAL practice problems written in the style and difficulty range of BMT - they are not actual or predicted BMT 2026 problems. Answers were checked by computation. Hide columns F-G to self-test.

#	Subject	Difficulty	Closest round	Problem	Answer	Solution sketch
1	Algebra	Easy	General	If x + 1/x = 3, compute x^5 + 1/x^5.	123	Let s_n = x^n + 1/x^n. Then s_(n+1) = 3 s_n - s_(n-1): 2, 3, 7, 18, 47, 123.
2	Algebra	Easy	General	Find the sum of all real solutions of |x^2 - 6x + 5| = 3.	12	x^2-6x+2=0 (two real roots, sum 6) and x^2-6x+8=0 (roots 2, 4, sum 6).
3	Algebra	Easy	General	The polynomial x^3 - 6x^2 + 5x + 7 has roots a, b, c. Compute (a+1)(b+1)(c+1).	5	P(x) = (x-a)(x-b)(x-c), so (a+1)(b+1)(c+1) = -P(-1) = -(-5) = 5.
4	Algebra	Medium	General / Guts	How many ordered pairs of integers (x, y) satisfy x^2 - y^2 = 2026?	0	(x-y)(x+y) has factors of equal parity, so the product is odd or a multiple of 4; 2026 = 2 mod 4.
5	Algebra	Medium	Algebra Focus	Positive reals x, y satisfy x + 2y = 6. Find the maximum of x*y^2.	8	AM-GM on x, y, y: (x+y+y)/3 = 2 >= (x y^2)^(1/3), equality at x = y = 2.
6	Algebra	Medium	General	Compute the sum of 1/(sqrt(k) + sqrt(k+1)) for k = 1 to 2024.	44	Each term equals sqrt(k+1) - sqrt(k); telescopes to sqrt(2025) - 1.
7	Algebra	Medium	Algebra Focus	A function satisfies f(x) + 2 f(1/x) = 3x for all nonzero x. Find f(2).	-1	Plug in x=2 and x=1/2: f(2)+2f(1/2)=6, f(1/2)+2f(2)=3/2. Solve.
8	Algebra	Hard	Algebra Focus	How many complex numbers z satisfy z^8 + z^4 + 1 = 0? What is the sum of their squares?	8 solutions; sum of squares 0	w = z^4 is a primitive cube root of unity (2 choices), each giving 4 values of z. Power sum p_2 = e_1^2 - 2 e_2 = 0 since the z^7 and z^6 coefficients vanish.
9	Algebra	Hard	Algebra Focus	Compute the sum of n^2 / 2^n for n = 1 to infinity.	6	Differentiate the geometric series twice, or use sum n x^n = x/(1-x)^2 and sum n^2 x^n = x(1+x)/(1-x)^3 at x = 1/2.
10	Geometry	Easy	General	Two circles of radii 3 and 12 are externally tangent. Find the length of a common external tangent segment between the points of tangency.	12	Length = 2*sqrt(r*R) = 2*sqrt(36).
11	Geometry	Easy	General	A regular hexagon has side length 2. Find the area of the triangle formed by three alternating vertices.	3*sqrt(3)	The triangle is equilateral with side 2*sqrt(3); area = (sqrt(3)/4)*12.
12	Geometry	Easy	General	Triangle ABC has AB = 7, AC = 9, BC = 8. Find the length of the median from A.	7	m_a^2 = (2*49 + 2*81 - 64)/4 = 49.
13	Geometry	Medium	General / Guts	Square ABCD has side 6. M and N are the midpoints of BC and CD. Lines AM and BN meet at P. Find the area of triangle ABP.	36/5	Coordinates: A(0,0), B(6,0), M(6,3), N(3,6). P = (24/5, 12/5); area = (1/2)(6)(12/5).
14	Geometry	Medium	Geometry Focus	A right triangle has legs 6 and 8. A semicircle has its diameter on the hypotenuse and is tangent to both legs. Find its radius.	24/7	Center (r, r) lies on x/6 + y/8 = 1, so r(1/6 + 1/8) = 1.
15	Geometry	Medium	Geometry Focus	A sphere is tangent to all 12 edges of a cube of side length 2. Find its volume.	8*sqrt(2)*pi/3	Radius = distance from center to an edge midpoint = sqrt(2); V = (4/3) pi (sqrt 2)^3.
16	Geometry	Hard	Geometry Focus	A triangle has side lengths 13, 14, 15. Find the distance between its incenter and circumcenter.	sqrt(65)/8	Area 84, r = 4, R = 65/8. Euler: OI^2 = R(R - 2r) = (65/8)(1/8).
17	Discrete	Easy	General	Find the last two digits of 7^2026.	49	7^4 = 2401 = 1 mod 100 and 2026 = 2 mod 4, so 7^2026 = 49 mod 100.
18	Discrete	Easy	General	Find the sum of the positive divisors of 2026.	3042	2026 = 2 * 1013 with 1013 prime; (1+2)(1+1013) = 3042.
19	Discrete	Easy	General	How many positive integers n <= 2026 are relatively prime to 2026?	1012	phi(2026) = 1 * 1012.
20	Discrete	Medium	General / Guts	In how many ways can the letters of BERKELEY be arranged so that no two E's are adjacent?	2400	Arrange B, R, K, L, Y in 5! ways, then choose 3 of the 6 gaps for the E's: 120 * 20.
21	Discrete	Medium	General	Three fair six-sided dice are rolled. What is the probability that the product is divisible by 4?	5/8	Complement: all odd (1/8), or exactly one die showing 2 or 6 with the others odd (3 * 1/3 * 1/4 = 1/4).
22	Discrete	Medium	Discrete Focus	How many subsets of {1, 2, ..., 10} (including the empty set) contain no two consecutive integers?	144	a_n = a_(n-1) + a_(n-2) with a_0 = 1, a_1 = 2 gives F_12 = 144.
23	Discrete	Medium	Discrete Focus	How many paths from (0,0) to (5,5) using unit steps right or up never go above the line y = x?	42	The 5th Catalan number: C(10,5)/6.
24	Discrete	Medium	Discrete Focus	Find the smallest positive integer n with n = 1 (mod 4), n = 2 (mod 5), n = 3 (mod 7).	17	Chinese Remainder Theorem; check 17 = 16+1 = 15+2 = 14+3.
25	Discrete	Hard	Discrete Focus	A fair coin is flipped until two consecutive heads appear. What is the expected number of flips?	6	States: E0 = 1 + E0/2 + E1/2, E1 = 1 + E0/2. Solve E0 = 6.
26	Calculus	Easy	Calculus Focus	Evaluate the limit as x -> 0 of (e^x - 1 - x) / (1 - cos x).	1	Both numerator and denominator are x^2/2 + higher-order terms.
27	Calculus	Easy	Calculus Focus	Evaluate the integral of x*sin(x) from 0 to pi.	pi	By parts: [-x cos x] from 0 to pi + integral of cos x = pi + 0.
28	Calculus	Easy	Calculus Focus	Find the area enclosed between y = x^2 and y = 2x + 3.	32/3	Curves meet at x = -1, 3; area = (3 - (-1))^3 / 6.
29	Calculus	Medium	Calculus Focus	Find the maximum value of f(x) = x * e^(-x^2/2) over all real x.	1/sqrt(e)	f'(x) = (1 - x^2) e^(-x^2/2) = 0 at x = 1.
30	Calculus	Medium	Calculus Focus	Let f(x) = x^x for x > 0. Compute f'(e).	2*e^e	f'(x) = x^x (ln x + 1).
31	Calculus	Hard	Calculus Focus	Evaluate the integral of 1/(1 + x^2)^2 from 0 to infinity.	pi/4	Substitute x = tan t: integral of cos^2 t from 0 to pi/2.
32	Estimation	Guts	Guts (final set)	Estimate the number of primes less than 2026. (Guts estimation problems give partial credit by closeness.)	306	n / ln n gives about 266; the better n/(ln n - 1) gives about 306.
33	Proof	Power	Power Round	Prove that in any group of 6 people, some 3 are mutual acquaintances or some 3 are mutual strangers.	(proof)	Pick a person; by pigeonhole 3 of the other 5 share the same relation to them. Examine the edges among those 3.
34	Proof	Power	Power Round	Prove that every tree on n >= 1 vertices has exactly n - 1 edges.	(proof)	Induct on n: a tree with n >= 2 vertices has a leaf (take the end of a longest path); delete it.
35	Proof	Power	Power Round	A length is 'constructible' if it can be obtained from 1 by finitely many +, -, *, / and square roots. Prove that the constructible numbers are closed under these operations, and show sqrt(2 + sqrt 3) is constructible.	(proof)	Mimics how a Power Round introduces a definition, then asks for closure properties and examples.

Summary

Counts update automatically if you add problems to the Practice Problems sheet.

Subject	# Problems	Easy	Medium	Hard
Algebra	9	3	4	2
Geometry	7	3	3	1
Discrete	9	3	5	1
Calculus	6	3	2	1
Estimation	1	0	0	0
Proof	3	0	0	0
Total	**35**	**12**	**14**	**5**

Real Past Papers

Work these under timed conditions; they are the best predictor of BMT 2026 difficulty.

Paper	Suggested use	Where
BMT 2025 (Nov 8, 2025) - all rounds	Full timed mock: most recent format	berkeley.mt/archives/bmt-2025
BMT 2024 (Nov 2, 2024) - all rounds	Second timed mock; Power Round on origami	berkeley.mt/archives/bmt-2024
BMT 2023 and earlier	Topic drills: pull problems 1-5 for warm-up, 8-10 for stretch	berkeley.mt/archives
Power Round archive 2012-2025	One Power Round per week as a team, written up properly	berkeley.mt/archives/bmt/power-round